Gasper Ndokaj

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EDUCATION

Vassar College Poughkeepsie, NY *May 2028*

Candidate for Bachelor of Arts

Major: Mathematics, minor in Continental Philosophy

Monsignor Farrell High School, Staten Island, New York *May 2024*

RELEVANT EXPERIENCE

Vassar College | *Research Assistant*, Poughkeepsie, NY *September 2025 – Present*

- Quantizing chaotic markers of dynamic diffraction time-series data of C. elegans locomotion in 3D using Recurrence Quantification Analysis, Lyapunov exponents, and Information Theory.
- Utilized techniques of nonlinear time-series analysis such as Takens's Embedding Theorem and Kolmogorov-Sinai entropy in MATLAB to show chaotic markers.

Polymath Jr. Research Program | *Researcher* *June 2026 - Present*

- Performed collaborative research on Binomial Edge Ideals under the supervision of Prof A. Seceleanu from the University of Nebraska-Lincoln.

Vassar College | *ACS Natural Sciences Assistant*, Poughkeepsie, NY *September 2025 – Present*

- Assist with maintaining The Laybourne Visualization Lab and Academic Computing Consultant (ACS) bookkeeping duties.
- Run software and licensing testing on computers in The Laybourne Visualization Lab.
- Maintain AirTable workspaces to track professor requests and collect workshop attendance and feedback.

INVITED TALKS

Hudson River Undergraduate Mathematics Conference | *Speaker,* Poughkeepsie, NY *April 2026 – April 2026*

- Gave a talk on Hutchinson's theorem and Iterated Function Systems in Fractal Geometry with Jay Chen and Henry Bonney at the HRUMC conference, under the guidance of Professor Natalie Frank of Vassar College.
- Gave intuition and background behind Hutchinson's theorem and focused on the topological and fractal geometric aspects of iterated function systems and Hutchinson's theorem for the existence and uniqueness of fractals.

Polymath Jr. Online Conference | *Speaker* *August 2026 – August 2026*

- Gave a talk on Binomial Edge Ideals alongside Shantanu Chakraborty of Clemson University, Ilya Galushkin of University of Pennsylvania, and Dragos Gilgor of Babes-Bolyai University for the ending conference for the Polymath Jr. Program.
- Focused on the Weak and Strong Lefschetz Properties for graded Artinian k-algebras and talked about findings from the Polymath Jr. program on their implications for binomial edge ideals of families of graphs.

SKILLS

Technical: LaTeX, Macaulay2, MATLAB, Linux, Command Line, Java, Vim, Python, Microsoft 365, AirTable, Proof-writing skills

Languages: English (Native), Albanian (Native/Fluent)

Interests: Abstract Algebra, Commutative Algebra, Algebraic Combinatorics, Combinatorics, Graph Theory, Nonlinear Dynamics, Chaos Theory, Information Theory, Mathematical Logic, Number Theory